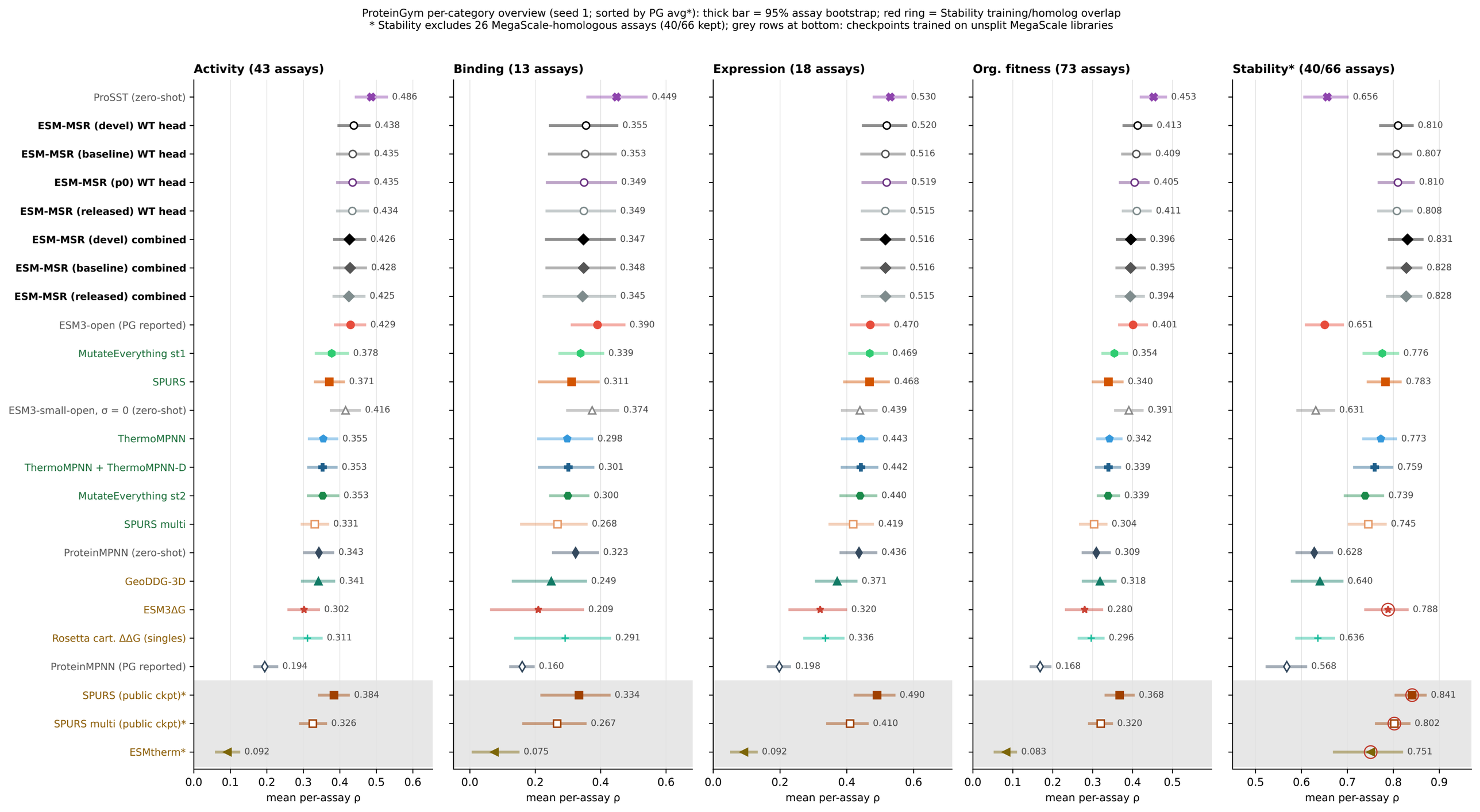
ProteinGym per-category overview (seed 1; sorted by PG avg*): thick bar = 95% assay bootstrap; red ring = Stability training/homolog overlap
* Stability excludes 26 MegaScale-homologous assays (40/66 kept); grey rows at bottom: checkpoints trained on unsplit MegaScale libraries
Activity (43 assays)
Binding (13 assays)
Expression (18 assays)
Org. fitness (73 assays)
Stability* (40/66 assays)
ProSST (zero-shot)
ESM-MSR (devel) WT head
ESM-MSR (baseline) WT head
ESM-MSR (p0) WT head
ESM-MSR (released) WT head
ESM-MSR (devel) combined
ESM-MSR (baseline) combined
ESM-MSR (released) combined
ESM3-open (PG reported)
MutateEverything st1
SPURS
ESM3-small-open, σ = 0 (zero-shot)
ThermoMPNN
ThermoMPNN + ThermoMPNN-D
MutateEverything st2
SPURS multi
ProteinMPNN (zero-shot)
GeoDDG-3D
ESM3ΔG
Rosetta cart. ΔΔG (singles)
ProteinMPNN (PG reported)
SPURS (public ckpt)*
SPURS multi (public ckpt)*
ESMtherm*
0.486 0.438 0.435 0.435 0.434 0.426 0.428 0.425 0.429 0.378 0.371 0.416 0.355 0.353 0.353 0.331 0.343 0.341 0.302 0.311 0.194 0.384 0.326 0.092
0.449 0.355 0.353 0.349 0.349 0.347 0.348 0.345 0.390 0.339 0.311 0.374 0.298 0.301 0.300 0.268 0.323 0.249 0.209 0.291 0.160 0.334 0.267 0.075
0.530 0.520 0.516 0.519 0.515 0.516 0.516 0.515 0.470 0.469 0.468 0.439 0.443 0.442 0.440 0.419 0.436 0.371 0.320 0.336 0.198 0.490 0.410 0.092
0.453 0.413 0.409 0.405 0.411 0.396 0.395 0.394 0.401 0.354 0.340 0.391 0.342 0.339 0.339 0.304 0.309 0.318 0.280 0.296 0.168 0.368 0.320 0.083
0.656 0.810 0.807 0.810 0.808 0.831 0.828 0.828 0.651 0.776 0.783 0.631 0.773 0.759 0.739 0.745 0.628 0.640 0.788 0.636 0.568 0.841 0.802 0.751
0.0 0.1 0.2 0.3 0.4 0.5 0.6
0.0 0.2 0.4 0.6
0.0 0.2 0.4 0.6
0.0 0.1 0.2 0.3 0.4 0.5
0.5 0.6 0.7 0.8 0.9
mean per-assay ρ